Akashkumar Jain

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Profile

MSc AI & Robotics candidate with hands-on experience in designing and developing AI and robotics systems.
Interested Areas: Physical AI, Robotics, Simulation, VLA / VLM / LLM , Computer Vision, Generative AI

Skills

- **Programming Languages:** Python , C++
- **ML Frameworks and Tools:** Data Pipelines, PyTorch, TensorFlow, Huggingface, Transformers, Large Language Model (LLMs), Vision Language Model (VLMs), Vision Language Action (VLAs), Model Fine-tuning (Full, LoRA & PEFT), Optimization & Quantization, Prompt Engineering, Experiment Design, Evaluation, Real-time Inference, Benchmarking
- **Robotics & Simulations:** ROS, SLAM, Kalman Filters, Sensor Fusion, Issac Sim, Issac Lab, MuJoCo, GYM, URDF, RCSS
- **Computer Vision:** Image & Video Processing, OpenCV, Faster-RCNN, Mask-RCNN, DETR, SAM2, Vision Transformer, Model Evaluation
- **Tools:** Git, Embedded Deployment (Jetson, Raspberry Pi), LeRobot, Linux, Jupyter Notebook, VS Code, Ollama, Docker, RunPod
- **Languages:** English (Fluent), German (A2-Limited working proficiency), Hindi (Native)

Education

MSc **University of Technology Nuremberg, DE** Oct 2023 – Mar 2026
AI and Robotics
- Advanced coursework and projects in AI, robotics, machine learning, Computer Vision, LLMs/VLMs, & Multimodal Foundation Models.

BE **KLE Technological University, IN** Aug 2016 – June 2020
Automation and Robotics
- CGPA: 8.02/10
- Specialization in Automation and Robotics.

Experience

Working Student: AI & Robotics, JARM.AI Schweinfurt, Germany
Nov 2025 – Present
- Developed and optimized Sim2Real and Real2Sim pipelines for robotic systems.
- Evaluated VLA model performance across simulated and real-world robotic environments.
- Validated model behavior and robustness under changing simulation and real-world conditions.

Machine Learning Freelancer Mumbai, India
Mar 2021 – Oct 2023
- Developed and deployed ML-based applications tailored to client needs.
  Key Projects:
  - Pickleball Tracking: Built real-time vision-based tracking systems using deep learning models.
  - Built Eye-Gaze Based Cursor Control for Assistive HCI.
  - Integrated ML models with ROS-based systems and IoT pipelines for perception-

driven applications.
  - Designed robotics simulation environments for manipulation tasks using PyBullet and OpenAI Gym.

Machine Learning Engineer, Nautilus Hearing Solutions — Hubli, India
Sep 2021 – Sep 2022

- Developed and deployed CNN-based models for Ear Health classification.
- Integrated machine learning pipelines into Android-based systems for real-world usage.

Research Intern, DRDO (CAIR) — Bangalore, India
Feb 2020 – May 2020

- Contributed to SLAM-based localization using fiducial markers (TagSLAM).
- Worked with ROS-based pipelines for map generation and pose estimation.

Master's Thesis

Improving Vision Transformer Efficiency Using Genetic Patch Pruning:

- Proposed a dynamic patch selection framework to improve Vision Transformer efficiency.
- Used a genetic algorithm to find optimal patch subsets and trained a Patch Selector to approximate this process.
- Adaptively selects informative patches per image and integrates with CLIP for improved efficiency.
- **Tools:** Python, PyTorch, Genetic Algorithms, CLIP, Vision-Language Models, MLPs, Vision Transformers

Projects (Master's)

Machine Learning

- **Object Detection Models:** Implemented Faster-RCNN, Mask RCNN, and DETR for visual object recognition..
  Tools: Python, PyTorch, OpenCV, Jupyter Notebook
- **Autonomous Driving:** Trained ML models on collected sensor and image data to simulate end-to-end self-driving capabilities in a virtual environment (Udacity simulator).
  Tools: Python, PyTorch, OpenCV, Jupyter Notebook, Udacity Simulator

Computer Vision

- **Captcha Solver:** Developed an end-to-end solution for recognizing alphanumeric CAPTCHAs with variable-length sequences, multiple degradation types, and challenging distortions.
- **Tools:** Python, Pillow, Data Augmentation, Data Generation, ResNet, CNN, Transformer Encoder and Parallel slot query decoder, PyTorch, hydra

Robotics

- **Playing Jenga with Franka:** Created a JengaEnv wrapper for gym, Programmed the Franka Emika 3 (FR3) robot to manipulate Jenga tower using precision actions in both simulation and real-world settings. Built a pipeline to collect data using simulation in MuJoCo.
- **Tools:** Python, MuJoCo, GYM, RCSS, CadQuery, XML, URDF

Large Language Models

- **UTN Chatbot:** Fine-tuned an LLM to answer student FAQs using LoRA and PEFT techniques and evaluated response accuracy and compiled technical report.
- **Tools:** Python, Huggingface, Transformers, PyTorch, LoRA, PEFT, Prompt Engineering

Multimodal Foundation Models

- **Optimized Tarsier:** Applied post-training quantization (INT8 and INT4) and reduced inference latency and memory footprint.
  Tools: Python, Tarsier2, Huggingface, Transformers, Bitsandbytes

- **Dash Grid:** Created and evaluated an LLM based agent to solve grid-world environments using the BabyAI platform, leveraging prompt engineering strategies.
  Tools: Python, BabyAI, Qwen2.5-7B-Instruct, gpt-4o-mini, Prompt Engineering

Personal

- **Video Background Remover:** Built a SAM2-based video background removal pipeline with API hosting and asynchronous processing using Celery.
  Tools: Python, SAM2, OpenCV, Fine-tunning, API Hosting, RunPod, Celery